

STRUCTURAL IDENTITY PRIMITIVE · CONFIDENTIAL

TopHash.

The Structural Identity Layer for the AI Era.

From bytes to structure. From hashing to proof-grade identity.

SYNOPSIS

TopHash is the missing structural identity primitive for software, science, and AI supply chains.

A training-free, theorem-backed structural fingerprint for any graph, an exact canonization layer with machine-auditable proof objects, and a counterfactual engine that certifies regime change.

ROUND

Series A

RAISE

\$20M

VALUATION

\$120M post

RUNWAY

24 months

01 · EXECUTIVE SUMMARY

TopHash in 90 seconds.

TopHash is a structural identity primitive. It does for graphs, molecules, dependency trees, transaction networks, and model architectures what SHA-256 did for bytes: produce a deterministic, comparable, and provably-attestable fingerprint. The core technology fuses persistent homology, spectral graph theory, and geometric statistics into a single self-tuned vector, then layers an exact canonization engine and a counterfactual perturbation algebra on top. The result is the first primitive that can answer three questions about any structured object at once: *what is it similar to, what is it identical to, and what would it become under admissible change.*

The market opportunity is structural. Eighty percent of enterprise data is fundamentally relational or graph-shaped, yet it is stored, indexed, and audited as if it were flat. Software supply-chain attacks grew forty percent year over year because SBOMs cannot prove structural identity between what was built and what was shipped. Drug discovery still wastes \$2.6B per approved molecule because molecular graph search is brute-force. AI regulators are demanding provenance attestation for models and datasets, and no primitive exists to provide it. TopHash dissolves this structural opacity tax across five independent \$1B+ verticals.

The team is led by ex-DeepMind, ex-Stripe, and ex-NSA researchers with a Chief Mathematician holding a PhD in algebraic topology from Oxford. Eight design partners are in private beta across cybersecurity, pharma, and fintech; three have signed letters of intent for the TopHashX Cloud API. Two published benchmarks beat Weisfeiler-Lehman kernels and graph neural network baselines on standard structure-only datasets (MUTAG, PROTEINS, NCI1). Eleven theorem families have been integrated into the reference implementation, providing the auditable proof substrate that machine-learning competitors cannot match.

We are raising \$20M Series A at a \$120M post-money valuation to ship TopHashX general availability, launch the TopHash Ω^∞ counterfactual engine, and scale enterprise GTM. Twenty-four-month runway takes the company to \$5M ARR, twenty-five paying enterprise logos, and the first Ω^∞ production deployments in drug discovery and AI governance. The outcome path is a \$1B+ primitive company — the structural identity layer for the AI era.

RAISE	VALUATION	TAM 2028	DESIGN PARTNERS	SIGNED LOIs
\$20M	\$120M post	\$80B+	8	3

This memo is structured in nine sections: thesis, problem, primitive, technology moat, market, go-to-market, competition, team & traction, and the ask.

02 · THE THESIS

Structure is the next frontier of identity.

Every major era of computing has been unlocked by a new identity primitive. The 1970s gave us cryptographic hashes for bytes — SHA, MD5, BLAKE — which made it possible to attest files at scale. The 2010s gave us learned embeddings for unstructured text — word2vec, BERT, GPT — which made it possible to search and compare natural language by meaning rather than by string. The 2020s have given us vector databases to operationalize those embeddings at planet scale. Each primitive created a multi-billion-dollar platform layer on top of it.

Structure is the next primitive layer, and it has no native identity yet. Graphs, molecules, dependency trees, transaction networks, model architectures, ontologies, schemas — all of these are first-class structural objects, and all of them are currently indexed, searched, and attested using primitives designed for bytes or text. The result is a structural opacity tax paid by every industry that handles structured data: unverifiable software supply chains, brute-force drug discovery, signature-based fraud detection, and AI models whose provenance cannot be attested.

TopHash is the missing primitive. It is deterministic, training-free, and theorem-backed. It produces three qualitatively different outputs from one engine: an approximate descriptor for retrieval (TopHash v3), an exact canonical form with machine-auditable proof objects (TopHashX), and a counterfactual invariant geometry that certifies what is fragile and what is critical (TopHash Ω^∞). The strategic implication is that TopHash is not a feature or a model — it is a primitive, like SHA-256, that other products are built on top of. Primitive companies compound differently than feature companies: every customer funds the same core, every vertical becomes a distribution channel for every other vertical, and every research advance strengthens the moat for everyone.

"TopHash is to structure what SHA-256 is to bytes." — that is the entire company in one sentence.

The remainder of this memo defends that claim: the problem and its cost (§3), the primitive and its three layers (§4-§5), the market and its five beachheads (§6), the GTM motion and the moat (§7-§8), the team and traction (§9), and the ask (§10).

03 · THE PROBLEM

Structure is the dark matter of data.

Modern data infrastructure hashes bytes, embeds text, and indexes rows. But eighty percent of enterprise data is fundamentally structural — software dependency graphs, molecular scaffolds, transaction networks, model architectures, ontologies, schemas. None of it has a native identity primitive. None of it is searchable by shape. None of it is provably attestable. The result is a structural opacity tax paid across every vertical that handles structured data, and the tax is large, growing, and concentrated in exactly the industries where regulators and capital allocators are now demanding proof.

THE COST, BY VERTICAL

VERTICAL	COST SIGNAL	ROOT CAUSE
Cybersecurity	\$4.45M avg cost per breach	Software supply-chain attacks grew 40% YoY (IBM 2024). SBOMs cannot prove structural identity between what was built, signed, and shipped — only that the byte-stream matches.
Drug Discovery	\$2.6B per approved molecule	90% of candidates fail in Phase I-II trials (DiMasi et al.). Molecular graph search is still brute-force; structural ADMET perturbation analysis does not exist at scale.
Financial Fraud	\$780K annual fraud loss per F1000	AFCC 2024. Transaction-graph anomaly detection is signature- and rule-based. Fraud rings restructure faster than rules can be authored.
AI Supply Chain	No provenance primitive exists	Regulators (EU AI Act, US Executive Order 14110) are demanding model and dataset attestation. No native structural fingerprint exists for model architectures or training corpora.
Software Supply Chain	12 months average SBOM drift	Endor Labs 2024. 51% of dependencies change quarterly. Without structural identity, drift detection requires byte-diff which produces massive false-positive rates.

These are not isolated costs. They share a single root cause: the absence of a structural identity primitive. TopHash dissolves the tax at the root rather than papering over each symptom with a vertical-specific band-aid. This is why a horizontal primitive company, rather than five vertical point solutions, is the right shape for the opportunity.

04 · THE PRIMITIVE

One engine. Three layers. Every structured object.

TopHash is not a single algorithm. It is a stack of three qualitatively distinct layers, each addressing a different question about a structured object. The layers compose: approximate retrieval narrows the search space, exact canonization proves identity, and the counterfactual engine certifies what would change under admissible perturbation. Customers can adopt one layer at a time, and pricing tiers map directly to layers — approximate is open-source, exact is usage-based cloud, counterfactual is enterprise.

LAYER 01	LAYER 02	LAYER 03
TopHash v3 Approximate structural fingerprint. Training-free, deterministic, fixed-size. Fuses persistence, spectral, and geometry views into a 52-dimensional (or 156-dimensional Ensemble) vector via a self-tuning quality-weighted engine. Searchable in milliseconds across billions of structured objects using standard ANN indexing. TRAINING-FREE · SEARCHABLE · 52D / 156D	TopHashX Exact canonization & proof. Five-stage pipeline: Search → Refine → Canon → Cert → ID. Produces canonical labeling, canonical serialization, SHA-256 structural identity digest, and machine-auditable proof objects containing the full refinement trace, witness logs, and validation records. Equality of canonical forms is isomorphism. EXACT MODE · PROOF OBJECTS · AUDITABLE	TopHash $\Omega\infty$ Counterfactual structural intelligence. Perturbation algebra generates admissible edits; response tensor measures per-view deltas; invariant-core / fragility-shell decomposition separates stable from sensitive channels; minimal-edit certificates identify the least-cost regime-flip. Backed by eleven theorem families. COUNTERFACTUAL · REGIME MAPS · THEOREM-BACKED

The strategic insight is that exactness does not come from adding more vector features. It comes from canonization: a deterministic procedure that maps every isomorphic graph to the same canonical form, then serializes that form and hashes the serialization. The vector stays approximate and fast; the canonical form is the source of truth; the proof object is the audit trail. This separation is what makes TopHash honest — approximate mode never claims to prove identity, and exact mode never has to be fast at retrieval. Competitors that try to do both in one learned model get neither property.

Formally: $TopHashX(G) = (a(G), C(G), H(C(G)), P(G))$ — approximate descriptor, canonical serialization, SHA-256 digest, and proof object. Correctness target: $C(G) = C(H)$ if and only if G and H are isomorphic, within the supported graph model.

05 · TECHNOLOGY MOAT

Eleven theorem families. One auditable proof object.

TopHash is defended not by a model checkpoint or a dataset moat, but by a theorem stack. Every certificate the primitive emits rests on a named mathematical result, which means every claim is independently auditable, reproducible across machines, and honest about its admissibility assumptions. Machine-learning competitors cannot match this without re-deriving the underlying mathematics, and even then they cannot eliminate the model-drift, retraining cost, and non-determinism that learned systems carry by default.

THE THEOREM STACK

#	THEOREM FAMILY	ROLE INSIDE TOPHASH
01	Persistence / Bottleneck Stability	Lipschitz bounds on topological summaries under bounded perturbation
02	Interleaving Distance	Module-level deformation distance, not just diagram-level
03	Gromov-Hausdorff Stability	Universal cross-domain metric-measure comparison
04	Cheeger Inequalities	Spectral shifts tied to true structural bottlenecks
05	Davis-Kahan Perturbation	Eigenvalue drift vs. eigenspace rotation separated
06	Eigenvalue Interlacing	Local edits imply constrained global spectral movement
07	Discrete Morse Theory	Critical edits, saddles, topological event structure
08	Conley Index / Morse Decomposition	Invariant basins, attractors, regime transition graphs
09	Optimal Transport / Kantorovich	Geometry over response distributions, not flat vectors
10	Graphon Compactness	Family-level asymptotic invariants, not just per-instance
11	Equivariant Persistence	Symmetry-aware — factors out benign relabelings

The practical consequence is that TopHash certificates carry their admissibility assumptions explicitly. A regulator, an auditor, or a third-party implementation can replay every proof step and check it against the published theorem. There is no black box, no model drift, no retraining pipeline, no GPU bill. The same input produces the same output forever, on every machine, in every jurisdiction. This is the substrate that cybersecurity, drug discovery, and AI governance customers need — and it is the moat that compound R&D deepens every quarter.

06 · MARKET

A horizontal primitive with five \$1B+ beachheads.

TopHash is horizontal infrastructure, but go-to-market is vertical. The primitive serves any industry that handles structured data, but the first five beachheads are chosen for acute pain, high willingness to pay, regulatory tailwinds, and a clear path to seven-figure ACVs. Total addressable market across the five beachheads exceeds \$80B by 2028, conservatively scoped to exclude adjacencies like knowledge graphs, ontologies, and graph-native analytics that would expand the surface further.

#	BEACHHEAD	TAM 2028	ICP & ENTRY PRODUCT
01	Cybersecurity & Software Supply Chain	\$12B	F500 CISOs, federal agencies, software distributors. TopHashX proof-grade SBOM identity, binary graph attestation, zero-trust structural certificates. Mandated by EU CRA, US Executive Order 14028, and NIST SP 800-218A.
02	Drug Discovery & Molecular Graph Search	\$18B	Top-20 pharma, biotech, CROs. TopHash v3 molecular similarity search at scale, TopHash Ω^∞ ADMET perturbation analysis, structural identity for IP and regulatory filing. Replaces brute-force Tanimoto search.
03	AI Supply Chain & Model Provenance	\$9B	AI labs, hyperscalers, regulators. AI Bill of Materials, model architecture fingerprints, dataset structural identity, training-corpus provenance attestation. Mandated by EU AI Act and US Executive Order 14110.
04	Financial Fraud & Transaction Graphs	\$14B	Banks, fintechs, payment networks. Real-time fraud-ring detection via TopHash v3 ANN over transaction graphs, counterparty structural identity, Ω^∞ regime-change alerts for emerging fraud topologies.
05	Data Infrastructure & Structural Indexing	\$27B	Graph DBs, lakehouses, vector DBs, data catalogs. TopHash v3 as the missing structural layer — native ANN over graphs integrated into Neo4j, Databricks, Snowflake, Pinecone. The long-term platform play.

Go-to-market sequence: cybersecurity and drug discovery first (acute pain, regulatory tailwinds, highest willingness to pay); AI supply chain and financial fraud second (regulated, high-volume, perfect for usage-based cloud pricing); data infrastructure last (long sales cycles, platform integrations, but the largest TAM and the most defensible long-term position). Each beachhead is independently a \$1B+ ARR outcome. Together, they compound into a platform.

07 · GO-TO-MARKET

Three-tier motion. Open-source funnel. Compounding moat.

TopHash goes to market as a three-tier product. Tier 1 is an open-source TopHash v3 SDK that drives developer mindshare and design-partner acquisition. Tier 2 is the TopHashX Cloud API, priced usage-based and designed for regulated industries that need exact-mode certificates and proof objects. Tier 3 is TopHash Ω^∞ Enterprise, deployed on-prem or in VPC, priced as annual contracts with dedicated research collaboration. The tiers map directly to the three technology layers, which keeps the story honest and the upsell motion natural.

TIER	PRODUCT	PRICING	BUYER & MOTION
01	TopHash v3 SDK · Open Source	\$0 · Apache 2.0	Developers, security researchers, data scientists. Bottom-up adoption via GitHub. Design-partner program with 8 launch partners.
02	TopHashX Cloud API	\$0.01 / 1K calls · \$0.10 / cert	Engineering teams in regulated industries. SOC 2, ISO 27001, FedRAMP-ready. Self-serve with enterprise contract ramp. ACV \$25K-\$250K.
03	TopHash Ω^∞ Enterprise	Custom · ACV \$500K-\$2M+	CISO, Chief Data Officer, Head of Drug Discovery. On-prem/VPC deployment. Dedicated research collaboration. Multi-year contracts.

UNIT ECONOMICS (24-MONTH HORIZON)

TopHashX Cloud gross margin targets 88% at scale (compute is sub-1ms per call, dominated by canonization cost on large graphs which is amortized via caching). Enterprise gross margin targets 72% (on-prem deployment support, dedicated research collaboration). Blended gross margin reaches 80% by month 24. CAC payback under 14 months for cloud, under 9 months for enterprise. Net Revenue Retention target 135% driven by usage expansion in cloud and by Ω^∞ upsell in enterprise.

Pricing logic: approximate mode is free because it is the funnel. Exact mode is usage-based because the value is per-certificate and the marginal cost is non-zero. Counterfactual mode is enterprise-priced because the value is strategic and the deployment is dedicated. This is the same three-tier shape that took HashiCorp, Snyk, and Postman from open-source projects to multi-billion-dollar platforms.

08 · COMPETITION

Why no existing player owns this.

The structural identity market is currently fragmented across four categories of competitor. None of them is a primitive. None of them offers exactness. None of them carries a theorem stack. The table below summarizes the positioning gap; the narrative underneath explains why the gap is durable rather than transitional.

CATEGORY	EXAMPLES	WHAT THEY DO	WHY TOPHASH WINS
Graph Neural Networks	PyG, DGL, GraphSAGE, GIN	Learned embeddings for downstream classification. Require training data, drift over time, non-deterministic across runs.	Training-free, deterministic, theorem-backed. No drift, no retraining, no GPU bill. Produces proof objects GNNs cannot.
Graph Kernels	Weisfeiler-Lehman, shortest-path, random walk	Hand-crafted similarity functions. Single-view (combinatorial only). Approximate, never exact.	Multi-view (persistence + spectral + geometry). Optional exact canonization. Self-tuning weights adapt across graph families.
Graph Databases	Neo4j, TigerGraph, Amazon Neptune	Storage and query layer for graphs. No identity primitive, no exactness, no proof layer.	TopHash is the layer above graph DBs — the missing structural index. Partnership, not competition.
SBOM / Attestation Tools	Sigstore, Syft, Anchore, Snyk	Byte-level hashing and signature verification. Cannot reason about structural identity or graph isomorphism.	TopHashX adds structural identity on top of byte identity. Complements, not competes with, existing SBOM tools.

The gap is durable for three reasons. First, the theorem stack cannot be replicated by a machine-learning team without re-deriving eleven mathematical results and shipping auditable proof objects — a multi-year research effort with no shortcut. Second, training-free determinism is a structural property, not a tuning choice; a learned model cannot be made deterministic without sacrificing its learning. Third, the cross-domain primitive dynamic means every TopHash customer funds the same core, so the R&D flywheel compounds across all five beachheads simultaneously. Competitors in any single vertical face a primitive company that is funded by four other verticals.

09 · TEAM & TRACTION

Built by the people who should be building this.

TEAM

The founding team combines the three disciplines a structural identity primitive company needs: deep mathematics, distributed systems engineering, and security-grade regulatory experience. No other team in the market combines all three at this depth.

ROLE	NAME & BACKGROUND
Co-founder & CEO	Dr. A. Vega — ex-DeepMind (AlphaFold team). PhD Algebraic Topology, Oxford. Published 24 papers on persistent homology and graph invariants.
Co-founder & CTO	M. Chen — ex-Stripe (Ledger team). 12 years building planet-scale distributed systems. Built payments infrastructure processing \$80B+ annual volume.
Chief Mathematician	J. Okafor — PhD Algebraic Topology, Oxford. Ex-NSA Research Directorate. Authority on canonical labeling algorithms and graph isomorphism complexity.
Advisor — Security	Former CISO of a F50 bank. Architect of US banking sector SBOM standards. Current board member of Cybersecurity & Infrastructure Security Agency advisory council.
Advisor — Drug Discovery	Former Head of Drug Discovery at a top-3 pharma. 22 years in computational chemistry. Led 4 molecules from candidate to FDA approval.

TRACTION (Q1 2026)

Eight design partners are in private beta across cybersecurity (3), pharma (3), and fintech (2). Three have signed letters of intent for the TopHashX Cloud API with average annual contract value of \$250K. Two benchmark papers have been published showing TopHash v3 outperforming Weisfeiler-Lehman subtree kernels and graph neural network baselines on standard structure-only datasets (MUTAG, PROTEINS, NCI1) — the only training-free method to do so. The Ω^∞ reference implementation has integrated eleven theorem families and is in early-access evaluation with two pharma partners for ADMET perturbation analysis.

DESIGN PARTNERS	SIGNED LOIs	BENCHMARKS PUBLISHED	THEOREM FAMILIES	AVG LOI ACV
8	3	2	11	\$250K

10 · THE ASK

\$20M Series A. 24 months to \$5M ARR.

We are raising \$20M Series A at a \$120M post-money valuation. The round is anchored by Crucible Governance Ltd (founder entity) with \$4M committed. We are seeking one to two institutional co-leads at \$6-8M each, with the remainder reserved for strategic angels and existing advisors. The capital funds twenty-four months of runway to TopHashX general availability, the launch of TopHash Ω^∞ v1, and the scaling of enterprise GTM from three design partners to twenty-five paying enterprise logos.

USE OF FUNDS

ALLOCATION	%	\$M	DETAILS
Engineering & Research	50%	\$10M	TopHashX GA ship, Ω^∞ v1 research and productization, infra scale to 10B+ indexed graphs, benchmark publications.
Go-to-Market	30%	\$6M	Enterprise sales hiring (6 AEs, 2 SEs), design-partner success team, developer relations for OSS funnel, marketing.
Research & IP	20%	\$4M	Theorem-stack publications, patent prosecution (8-12 filings), academic collaborations with Oxford, MIT, Stanford.

18-MONTH MILESTONES

WHEN	MILESTONE
Month 6	TopHashX Cloud GA. SOC 2 Type II audit complete. First 5 paying cloud customers live.
Month 9	TopHash v3 OSS 1.0. 10K GitHub stars. 25K weekly PyPI installs. First integration with a major graph database (Neo4j or TigerGraph).
Month 12	TopHash Ω^∞ v1 early-access with 2 pharma partners. First FDA-facing structural identity case study published.
Month 15	\$2M ARR run rate. 15 paying enterprise logos. FedRAMP-ready certification package submitted.
Month 18	\$5M ARR run rate. 25 paying enterprise logos. Ω^∞ v1 GA. Series B readiness deck circulated.

KEY RISKS & MITIGATIONS

RISK	DESCRIPTION	MITIGATION
Canonization scaling	Exact mode on very large graphs (10M+ nodes) is computationally expensive.	Phase exact mode behind Refine layer; use approximate mode for retrieval. Cache canonical forms. Long-term: sparse and parallel canonization research.

Competitive response	Hyperscalers or graph DB vendors could attempt to build a competing primitive.	Theorem stack is multi-year research; cross-domain flywheel compounds R&D; faster than any single-vertical competitor can match. Patent prosecution underway.
Regulatory timing	AI supply-chain and SBOM regulations could slip, delaying beachhead-3 revenue.	Cybersecurity and drug discovery beachheads are independent of AI regulation timing. AI supply chain is upside, not base case.
Talent concentration	Chief Mathematician and CEO are singular; loss would slow research velocity.	Oxford and MIT academic collaborations create a deep talent funnel. Equity grants and publication policy make TopHash a magnet for topological talent.

TopHash is the structural identity layer for the AI era — and the next \$1B+ primitive company. We are looking for institutional partners who understand primitive companies, who can hold a ten-year horizon, and who can help us navigate regulated verticals. We would love to continue the conversation.

Contact: founders@tophash.io · tophash.io · Crucible Governance Ltd, 2026